RAMCO INSTITUTE OF TECHNOLOGY

Department of Mechanical Engineering

Academic Year 2020 – 2021 (Odd Semester)

Innovative practice(s) description

Course code and title: ME8792 Power Plant Engineering

Name of the Instructor(s): Dr.V.Sivakumar, ASCP/Mechanical
Mr.M.Ashok Kumar, AP(SG)/Mechanical

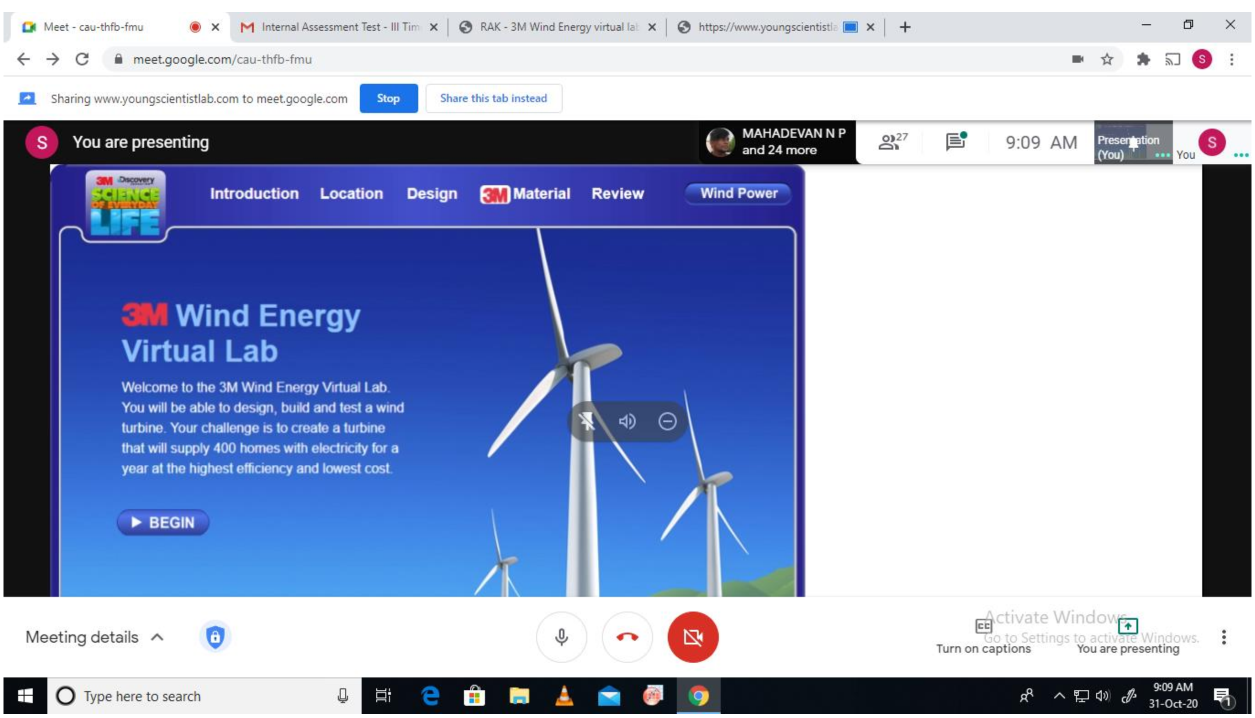
Date & Time: 31.10.2020 & 9.00 AM to 9:50 AM

Virtual Laboratory demonstration on 3M Wind Energy

For the topic "Site selection criteria for Wind farm installation", to make the students to understand better, the virtual laboratory ***"3M Wind Energy"*** developed by Young Scientist Laboratory was used. The tool enabled to design a wind farm in three different locations:

a. Off – shore b. On – shore plains c.Hills

The virtual lab also enabled to select the other design parameters including blade length, blade pitch, blade twist, tip shape, airfoil shape, height of turbine and 3M materials. Selection of these parameters at different locations helps to understand the impact of selection criteria in installing a wind turbine.



Screenshot of the lab used

To make the students to understand better, an assignment was given in which the students have to submit a report by practicing the experiment. The practicing was in such a way that the students have to keep 7 parameters fixed and alter any one parameter to generate three different results. This makes them to have a better clarity in selection criteria.

This activity helps in attaining the following CO – PO mapping:

Course Outcome / Programme Outcome	**PO1**	**PO2**	**PO3**	**PO4**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO12**
CO4 – Student will be able to explain the principle/construction and working of hydroelectric and various non-conventional power plants.	3.0	2.2	1.0	2.3	3.0	2.3	2.0	2.0	1.6	3.0